TARGET

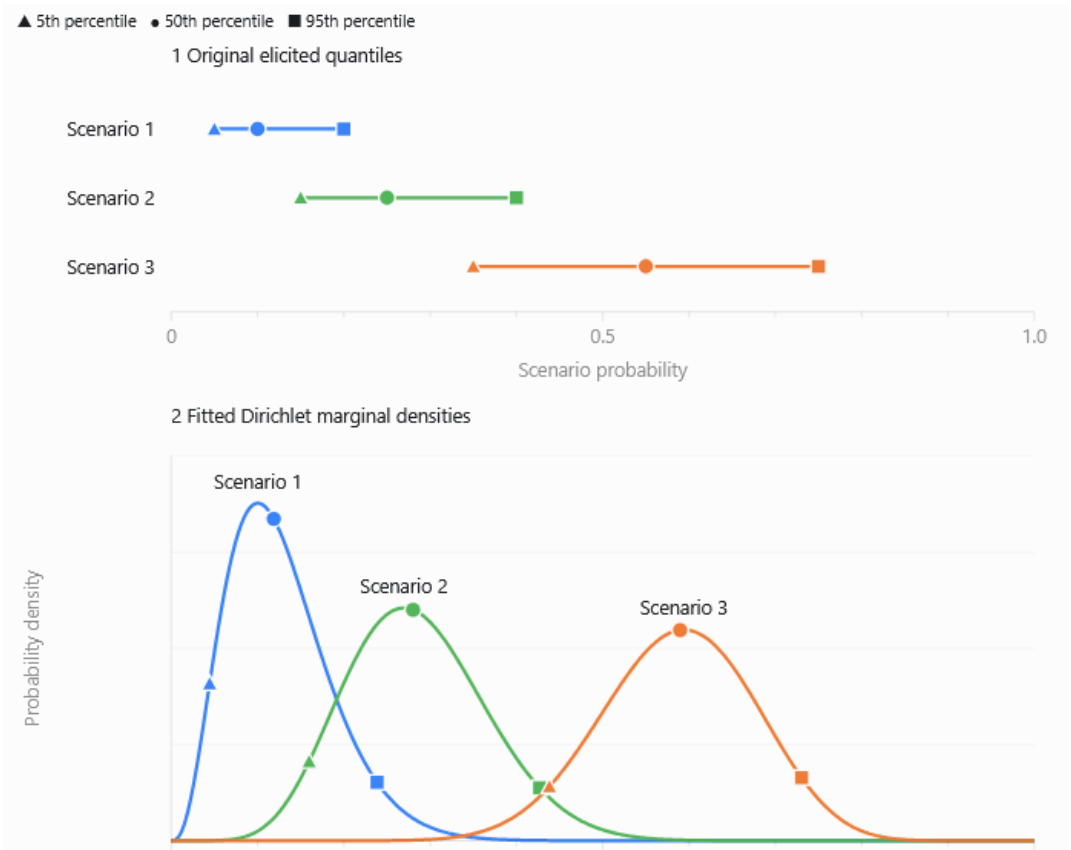
First Name | Last Name | email

1. Persistence of the southwestern Hokkaido tectonic regime

Let TTR denote the additional time from the assessment date until the present regional tectonic regime affecting the defined southwestern Hokkaido RES area first undergoes a substantive and sustained reorganization. Such a reorganization may involve one or more of the following: (i) a sustained, significant change in the rate or direction of the specified relative plate motion; (ii) a sustained redistribution of deformation among the principal tectonic structures or zones; or (iii) a substantive change in the geometry or location of the regional tectonic regime. Transient earthquake-cycle, postseismic or other short-term geodetic changes should not be treated as termination of the regime. Please provide your 5th-, 50th- and 95th-percentile judgments for TTR, expressed in units of 10 ka. (units of 10 ka)

5%ile | 50%ile | 95%ile

Also state directly your probability that TTR will be at least 1 Myr (Pr [TTR ≥ 1 Myr]) :



2.1. Long-term uplift along the Shakotan Peninsula – Hidden active faults within 5 k

Modest regional uplift at rates of up to 0.5 m per thousand years, as indicated by the elevation of former interglacial shorelines (MIS 5 – 7), occur along the southwest coast of the Shakotan Peninsula in SW Hokkaido (Amano et al., 2018; Nuclear Power Plant Environmental Improvement Organisation, 2024). This uplift is distant from currently mapped active faults (> 10□30 km distant) and active volcanoes (> 30□65 km). The cumulative effect of uplift over a few thousand years or longer could result in eventual daylighting of a repository via uplift and/or erosion, or in significant changes to the oxidizing/reducing environment around a repository over long timescales. For example, an active hidden fault could entail metre scale cumulative displacement in around 10,000 years, for overall uplift of 50 metres in about 100ka. Two other mechanisms are possible continuous processes, e.g., magma intrusion or isostatic adjustment, each with potential for long-term cumulative uplift effects. Our goal is to characterize our current knowledge of this uplift scenario by encoding individual expert's 5th, 50th, 95th percentiles and pooling these judgments to generate a synthesis assessment of uncertainties relating to these potential contributions to uplift (see illustrative example plot). Three probable situations are identified below that could contribute to long-term uplift along the Shakotan Peninsula. (If you consider other mechanisms would be appropriate to consider in this context in future project discussions, please briefly note them in the text box below). Please ascribe your median weights for the relative contributions of each mechanism and your 5th and 95th percentiles for each weight, all values must be within the relative weight interval [0..1]. Notes: Median and quantile weights = 0 indicate you judge that a mechanism makes no contribution to Shakotan Peninsula uplift. Otherwise, your three median fractional weights should sum close to 1.0 (strictly, the means sum to 1.0). Your marginal 5th and 95th percentiles are not simultaneous realizations of the process but a way of expressing your uncertainty around your median weight; these marginal percentile weights will not sum to 1.0, but each value must lie within interval [0..1]. ( )

5%ile	50%ile	95%ile



2.2. Long-term uplift along the Shakotan Peninsula – Magma intrusion < 15 km fro

Modest regional uplift at rates of up to 0.5 m per thousand years, as indicated by the elevation of former interglacial shorelines (MIS 5 – 7), occur along the southwest coast of the Shakotan Peninsula in SW Hokkaido (Amano et al., 2018; Nuclear Power Plant Environmental Improvement Organisation, 2024). This uplift is distant from currently mapped active faults (> 10□30 km distant) and active volcanoes (> 30□65 km). The cumulative effect of uplift over a few thousand years or longer could result in eventual daylighting of a repository via uplift and/or erosion, or in significant changes to the oxidizing/reducing environment around a repository over long timescales. For example, an active hidden fault could entail metre scale cumulative displacement in around 10,000 years, for overall uplift of 50 metres in about 100ka. Two other mechanisms are possible continuous processes, e.g., magma intrusion or isostatic adjustment, each with potential for long-term cumulative uplift effects. Our goal is to characterize our current knowledge of this uplift scenario by encoding individual expert's 5th, 50th, 95th percentiles and pooling these judgments to generate a synthesis assessment of uncertainties relating to these potential contributions to uplift (see illustrative example plot). Three probable situations are identified below that could contribute to long-term uplift along the Shakotan Peninsula. (If you consider other mechanisms would be appropriate to consider in this context in future project discussions, please briefly note them in the text box below). Please ascribe your median weights for the relative contributions of each mechanism and your 5th and 95th percentiles for each weight, all values must be within the relative weight interval [0..1]. Notes: Median and quantile weights = 0 indicate you judge that a mechanism makes no contribution to Shakotan Peninsula uplift. Otherwise, your three median fractional weights should sum close to 1.0 (strictly, the means sum to 1.0). Your marginal 5th and 95th percentiles are not simultaneous realizations of the process but a way of expressing your uncertainty around your median weight; these marginal percentile weights will not sum to 1.0, but each value must lie within interval [0..1]. ( )

5%ile	50%ile	95%ile



2.3. Long-term uplift along the Shakotan Peninsula – Local isostatic response to se

Modest regional uplift at rates of up to 0.5 m per thousand years, as indicated by the elevation of former interglacial shorelines (MIS 5 – 7), occur along the southwest coast of the Shakotan Peninsula in SW Hokkaido (Amano et al., 2018; Nuclear Power Plant Environmental Improvement Organisation, 2024). This uplift is distant from currently mapped active faults (> 10□30 km distant) and active volcanoes (> 30□65 km). The cumulative effect of uplift over a few thousand years or longer could result in eventual daylighting of a repository via uplift and/or erosion, or in significant changes to the oxidizing/reducing environment around a repository over long timescales. For example, an active hidden fault could entail metre scale cumulative displacement in around 10,000 years, for overall uplift of 50 metres in about 100ka. Two other mechanisms are possible continuous processes, e.g., magma intrusion or isostatic adjustment, each with potential for long-term cumulative uplift effects. Our goal is to characterize our current knowledge of this uplift scenario by encoding individual expert's 5th, 50th, 95th percentiles and pooling these judgments to generate a synthesis assessment of uncertainties relating to these potential contributions to uplift (see illustrative example plot). Three probable situations are identified below that could contribute to long-term uplift along the Shakotan Peninsula. (If you consider other mechanisms would be appropriate to consider in this context in future project discussions, please briefly note them in the text box below). Please ascribe your median weights for the relative contributions of each mechanism and your 5th and 95th percentiles for each weight, all values must be within the relative weight interval [0..1]. Notes: Median and quantile weights = 0 indicate you judge that a mechanism makes no contribution to Shakotan Peninsula uplift. Otherwise, your three median fractional weights should sum close to 1.0 (strictly, the means sum to 1.0). Your marginal 5th and 95th percentiles are not simultaneous realizations of the process but a way of expressing your uncertainty around your median weight; these marginal percentile weights will not sum to 1.0, but each value must lie within interval [0..1]. ( )

5%ile	50%ile	95%ile

Use this space to note any other uplift mechanisms which could affect the Shakotan Peninsula and should be considered in future NUMO-TOPAZ project work: